

Opisthobranchs of the Gulf of Mannar Biosphere Reserve, Tamil Nadu, India

J. S. YOGESH KUMAR, C. R. SREERAJ AND R. SORNARAJ*

Zoological Survey of India, Andaman and Nicobar Regional Centre, National Coral Reef Research Institute
Haddo, Port Blair- 744 102, Andaman and Nicobar Islands, India

*Department of Zoology, Kamaraj College, Tuticorin, Tamil Nadu, India
e-mail: crsreeraj@gmail.com

ABSTRACT

The Gulf of Mannar, located in the south-east coast of India with an area of 10,500 km² (1. 8° 47' - 9° 15' N and 1. 78° 12' - 79° 14' E), is a major fishing zone in the Indian waters. Several fisher communities make their livelihood by exploiting finfishes and shellfishes as well as marine algae and other resources from this area. Extensive survey for sea slugs was conducted during 2008 in the sub-tidal regions of Vembar and Tuticorin groups of islands in the Gulf of Mannar Biosphere Reserve using SCUBA. The present study brings out 14 species of nudibranchs new to this region. Also one species each under both cephalaspidean and sacoglossan opisthobranchs described in the paper are new to this region. A total of 20 species of opisthobranchs were recorded during the study. Literature reveals that 47 species of nudibranchs have been recorded previously from the Gulf of Mannar region.

Keywords: Checklist, Gulf of Mannar, New record, Nudibranch, Opisthobranch

Introduction

The first available report of opisthobranchiate fauna from India is the collection of nudibranchiate molluscs made by Walter Eliot (Alder and Hancock, 1864) from Waltair which dealt with 42 species belonging to 10 families each of which, 30 species and 4 genera were newly described. The first available report on nudibranchs from the Gulf of Mannar is by Farran (1905), based on the collection made by Herdman from Ceylon. This collection include specimens collected from the Palk Strait also, but the particular locality details are not available and so the species in the list are not authenticated as that of Gulf of Mannar. Subsequently, Eliot (1906) published a report on the nudibranchs of southern India and Ceylon (Sri Lanka) which includes collections by Elliot near Visakhapatnam, collections and drawings by Keelart, and a collection named, Madras (or India) 1867 by Walter Eliot. Subsequently, O'donoghue in 1932 published reports on the nudibranchs from the Gulf of Mannar. There were many works by Indian researchers thereafter from this area. Most of the taxonomical works were carried out by Gravally (1942); Rao and Alagarswami (1960, 1961); Satyamurti Rao (1962a, b; 1967a, b; 1968; 1973); Rao and Rao (1963); Rao (1968); Reddiah (1970); Rao and Krishnakumari (1973; 1974) and Valdes *et al.* (1999). In 2001, chemo-ecological studies on the opisthobranchiate fauna was also published from this area (Fontana *et al.*, 2001).

Thorough search of literature reveals 47 species of nudibranchs from the Gulf of Mannar, which is

considerably a very poor representation of the richly diversified sea slugs of the region compared to 3400 species estimated to occur in the Indo-West Pacific Province. The present paper reports the occurrence of 20 species of opisthobranchs from the Gulf of Mannar Biosphere Reserve (GOMBR), of which 14 species of nudibranchs were new record to this area. Also one species each under cephalaspidean and sacoglossan opisthobranchs are new to this marine reserve.

Materials and methods

An extensive survey of sea slugs was conducted during 2008 in various regions of the Gulf of Mannar Biosphere Reserve. SCUBA gear was used for surveying the subtidal regions. The surveys were carried out in the Vembar group of islands (Upputhanni, Puluvinichalli and Nallathanni) and Tuticorin group of islands (Van, Kaswary, Kariyachalli and Vilanguchalli) in the Gulf of Mannar. The patchy reef area other than the island periphery locally known as "Mainland" was also surveyed. Most of the previous works from the Gulf of Mannar were based on the Mandapam group of islands (Musal, Manoli, Manoliputti, Poomarichan, Pullivasal, Krusadai and Shingle) and Kezhakkarakai group of islands (Anaipar, Valimunai, Poovarasani Patti, Appa, Thalairi, Valai and Mulli). The nudibranchs were observed and photographed underwater using Nikon D50 and Canon Powershot S45 cameras with underwater housing. Specimens were not collected as the study area is a Biosphere Reserve. Characters used in this study for the identification of species are: general

shape and body profile, colour and pattern, morphology of notal tubercles, ridges, and the mantle margin, rhinophoral colour; morphology of foot, foot sole and oral tentacles.

Results and discussion

A total of 20 species of sea slugs belonging to six families and three orders were recorded during this study in the Gulf of Mannar Biosphere Reserve. The morphological features of presently observed nudibranchs are given below:

ORDER : CEPHALASPIDEA Fischer, 1883
 SUPER-FAMILY : PHILINOIDEA Gray, J. E., 1850
 FAMILY : AGLAJIDAE Pilsbry, 1895 (1847)

Philinopsis speciosa Pease, 1860

Description: The ground colour of the animal ranges from light translucent brown to an opaque dark brown or even black. There is a pair of yellow to orange-brown parallel lines on the head shield on either side of the midline, and sometimes traces of a second line on each side, further out, similar to those sometimes present in *P. cyanea*. The white spots range from quite small to very large, the large white markings being somewhat broken by brown speckling. The margin of the parapodia is colored with black, blue and orange-yellow markings, which can sometimes form lines. The foot appears to be of a similar colour to the dorsal surface (Fig. 1a).

Material observed: Van Island (08° 49' 45.8", 078° 13' 10.2"); Maximum length of the crawling animal: ~2.5cm; depth: 12 m; sandy area covered with algal mat.

Distribution: Hawaii, Indonesia, India

Remarks: New record to Gulf of Mannar Biosphere Reserve

ORDER : SACOGLOSSA (von Ihering, 1876)
 SUPER-FAMILY : ELYSIOIDEA Forbes and Hanley, 1851
 FAMILY : ELYSIIDAE Forbes and Hanley, 1851

Elysia grandifolia Kelaart, 1858

Description: Described by Kelaart from Ceylon (Sri Lanka), it is very similar in colour to *Elysia ornata*. It grows to a very large size; the observed specimens were more than 12 cm in size. The animal is usually translucent green colour with a black parapodial margin and a submarginal yellow or orange band. The parapodia are large and very thin, and the black and orange bands at the edge of the parapodia are not separated by white. There are also numerous black and white dots all over the body. This species was found feeding on algae.

Material observed: Puluvinichalli Island (09° 06' 10.21", 078° 32' 20.5"); maximum length of the crawling animal: ~8-13 cm; depth: 15 m; coral reef area with rocks covered by algae (Fig. 1b).

Distribution: Indian Ocean

Remarks: Reported earlier from this area (Satyamurti, 1952). Animals which resembled both *E. grandifolia* and *E. ornata* were observed.

Syphonota geographica (Adams and Reeve, 1850)

Description: This circumtropical species can be distinguished from species of *Aplysia* by the position of the rhinophores, close together and set back between the parapodial lobes. They swim by flapping the parapodia. The name *geographica* refers to the map-like nature of the colour pattern and *scripta* also refers to the colour pattern. Animal almost blend with the background flora (Fig. 1c).

Material observed: Puluvinichalli Island (09° 06' 10.21", 078° 32' 20.5"); maximum length of the crawling animal: ~13 cm; depth: 15 m; sandy area covered with algal mat.

Distribution: Indo-West Pacific and Atlantic

Remarks: New record to Gulf of Mannar Biosphere Reserve

ORDER : NUDIBRANCHIA Blainville, 1814
 SUB-ORDER : DORIDINA Odhner, 1934
 FAMILY : DORIDIDAE Rafinesque, 1815

Jorunna funebris (Kelaart, 1858)

Description: Body oblong, convex, of a waxy-white colour, and spotted with black. Mantle coriaceous, granular, of an ivory white colour and ornamented with jet black spotted circles and half rings or imperfect annular spotted figures. Dorsal tentacles large, clavate; apex black, laminated, without sheaths. Oral tentacles linear, white, tips black. Branchial plumes six, large and drooping, tripinnate, white and shaded lavender grey; midribs of a dark brown colour. Foot waxy white, spotted irregularly on the margin of the edges with small and large linear spots (Fig. 1d).

Material observed: Nallathanni Island (09° 06' 04.3", 078° 34' 39.7"); maximum length of the crawling animal: ~5 cm; depth: 20 m; reef area with rocks in between.

Distribution: Widespread in Tropical Indo-West Pacific.

Remarks: Reported from this area by Fontana *et al.* (2001)

ORDER : NUDIBRANCHIA Blainville, 1814
 SUB-ORDER : DORIDINA Odhner, 1934
 FAMILY : CHROMODORIDIDAE Bergh, 1891

Chromodoris fidelis (Kelaart, 1858)

Description: The mantle is creamy white with a broad orange band around the edge and a thin wine red line between the white and the orange. On the inner edge of the orange band, there are usually a series of tongue-like patches running into the white region. In some cases they are almost all wine red while in others there is only a slight

thickening of the red band. Equally characteristic are the black rhinophores and gills and the white foot (Fig. 1c).

Material observed: "Mainland" (08° 49' 28.8", 078° 14' 28.62"); maximum length of the crawling animal: ~4 cm; depth: 16 m; sandy area with seaweeds and seagrass.

Distribution: Common in tropical Indo-West Pacific.

Remarks: New record to Gulf of Mannar Biosphere Reserve

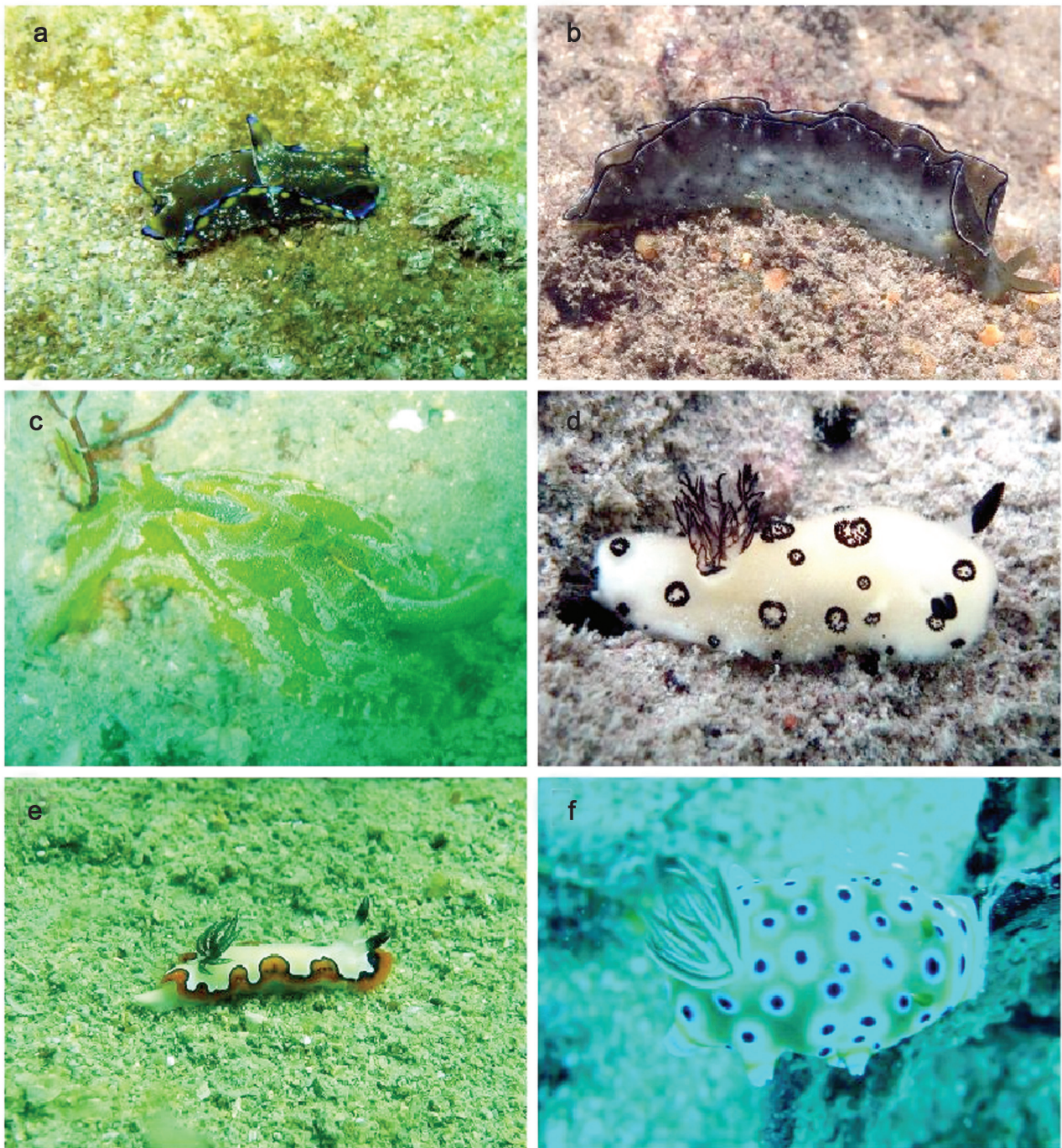


Fig. 1. Underwater photographs of the species observed (a) *Philinopsis speciosa* (b) *Elysia grandifolia* (c) *Syphonota geographica* (d) *Jorunna funebris* (e) *Chromodoris fidelis* (f) *Chromodoris geminus*

Chromodoris geminus Rudman, 1987

Description: This is one of the species included in the *Risbecia tryoni* colour group as per Rudman (1987). Other species which include are *Risbecia tryoni*, *Chromodoris kuniei* and *Chromodoris leopardus*. They have large purple or purple-brown spots or marks, usually ringed with white, an often reticulate brownish background, and a purple border.

Chromodoris geminus differs in having four colour bands around the mantle edge, an outermost white, then translucent greyish purple, and then white followed by yellow. Others have a distinct purple line at the border, narrow in *C. leopardus* and *R. tryoni*, but broad in *C. kuniei* (Fig. 1f).

Material observed: Kariyachalli Island (08° 57' 02.6", 078° 15' 18.8"); maximum length of the crawling animal: ~7

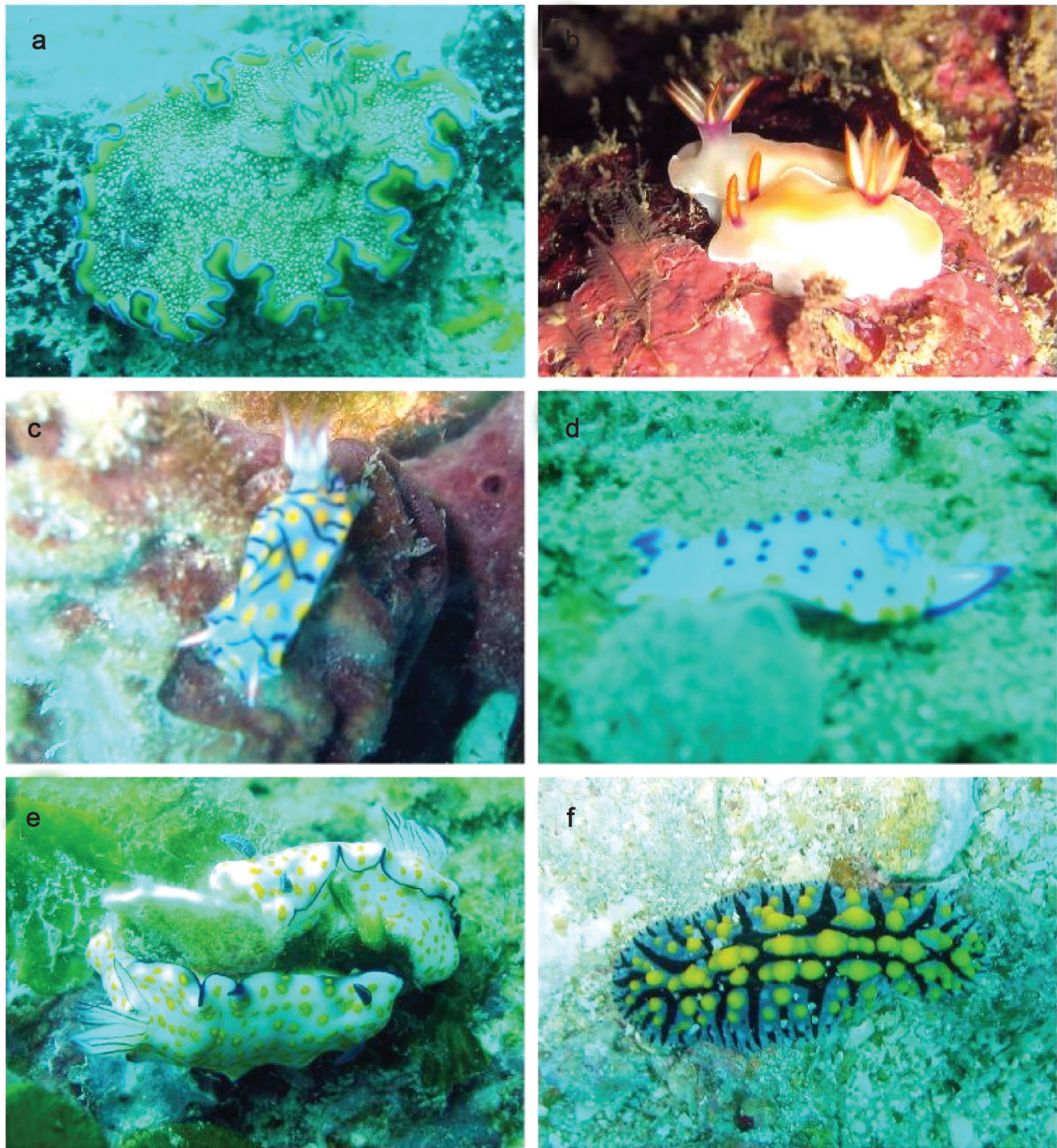


Fig. 2. Underwater photographs of the species observed (a) *Glossodoris cincta* (b) *Hypselodoris bullocki* (c) *Hypselodoris nigrostriata* (d) *Mexichromis multituberculata* (e) *Risbecia pulchella* (f) *Fryeria marindica*

cm; depth: 20 m; reef area with sandy patches in between.

Distribution: Tropical Indian Ocean (Indonesia, Thailand, Red Sea, Maldives, South Africa and India)

Remarks: New record to Gulf of Mannar Biosphere Reserve

Glossodoris cincta (Bergh, 1888)

Description: This widespread Indo-West Pacific species seems to have three regional colour forms. The species occur in this area has a bluish outer band on the mantle edge but the inner black and yellow lines tend to merge into a dull khaki band. The dull khaki colour of the mantle extends to the rhinophores and the gills. The rhinophoral tip is also similar to the mantle edge colour combination (Fig. 2a).

Material observed: Kariyachalli Island (08° 57' 02.6", 078° 15' 18.8"); maximum length of the crawling animal: ~2.5 cm; depth: 10 m; reef area; "Mainland" (09° 07' 06.9", 078° 40' 49.1"); maximum length of the crawling animal: ~5 cm; depth: 15 m; algae and seaweed covered area. Vilankuchalli Island (08° 56' 09.4", 078° 16' 05.7"); maximum length of the crawling animal: ~6 cm; depth: 13 m; sandy area with sponges and soft corals, the animal was seen feeding on a sponge (*Ircinia* sp.).

Distribution: Throughout Indo-West Pacific.

Remarks: New record to Gulf of Mannar Biosphere Reserve

Hypselodoris bullocki (Collingwood, 1881)

Description: It ranges in colour from a pale straw, or even white background to a deep purplish pink. The species observed was bluish in color. There is a thin opaque white line at the mantle border. Typically the gills and rhinophores are yellow or orange with a basal pink or purplish band. The mantle is smooth and without any colour or other ornamentations. Rhinophore is yellow to orange with a dark blue to purple base (Fig. 2b).

Material observed: "Mainland" (09° 00' 25.4", 078° 24' 15.3"; 08° 54' 37.1", 078° 16' 38.5"); maximum length of the crawling animal: 4 – 6 cm; depth: 15 m; rocky area covered with sand and algal mat.

Distribution: Tropical western Pacific and eastern Indian Ocean.

Remarks: New record to Gulf of Mannar Biosphere Reserve

Hypselodoris nigrostriata (Eliot, 1904)

Description: This species is characterized by dark blackish lines, usually diagonal, with yellow patches between. The background colour of the body is bluish; the rhinophores are orange red and the gills, triangular in cross-section and are edged with the same colour. There is a typical 'W' shaped black line on the cephalic region (fig. 2c).

Material observed: "Mainland" (09° 07' 06.9", 078° 40' 49.1"); maximum length of the crawling animal: ~3.5 cm; depth: 15 m; rocky area covered with seaweeds.

Distribution: Throughout Indian Ocean

Remarks: New record to Gulf of Mannar Biosphere Reserve

Mexichromis multituberculata (Baba, 1953)

Description: *M. multituberculata* was found to differ from the other western Pacific species of *Mexichromis* in having purple spots around the edge of the mantle rather than yellow or orange. The mantle has scattered purple-tipped tubercles. However, many reports were published where the animals were with reddish, purple, blue or orange spots around the mantle edge. The mantle is waxy white in colour, with blue tipped tubercles (Fig. 2d).

Material observed: Van Island (08° 50' 07.4", 078° 12' 32.1"); maximum length of the crawling animal: ~2.5cm; depth: 20 m; sandy area covered with algal mat and seaweeds.

Distribution: Seto, Kii, Japan and Hainan Island, Hong Kong, Sulawesi, Thailand, Papua New Guinea and India.

Remarks: New record to Gulf of Mannar Biosphere Reserve

Risbecia pulchella (Ruppell and Leuckart, 1828)

Description: Opaque creamy white mantle, marked by a bluish purple tinge in irregular areas on the mantle and sides of the foot. The dorsal surface of the mantle and the foot is covered with yellowish orange spots of irregular shape of different sizes. The mantle margin is fringed with a thin border of bright bluish violet. The rhinophores are blue in colour. The gills are simple and creamy yellow in colour with an orange coloured line. The gills are often branched and wave rhythmically from side to side. The front part of mantle has an inward curving typical to this species (Fig. 2e).

Material observed: Puluvinichalli Island (09° 06' 16.3", 078° 32' 09.6"); maximum length of the crawling animal: ~9 cm; depth: 18 m; rocky area covered with algal mat and seaweeds.

Distribution: Indian Ocean. Relatively common in the Red Sea and East Africa.

Remarks: New record to Gulf of Mannar Biosphere Reserve

ORDER : NUDIBRANCHIA Blainville, 1814

SUB-ORDER : DORIDINA Odhner, 1934

FAMILY : PHYLLIDIIDAE Rafinesque, 1815

Fryeria marindica (Yonow and Hayward, 1991)

Description: *F. marindica* is predominately bluish grey in colour with two longitudinal black bands running the length

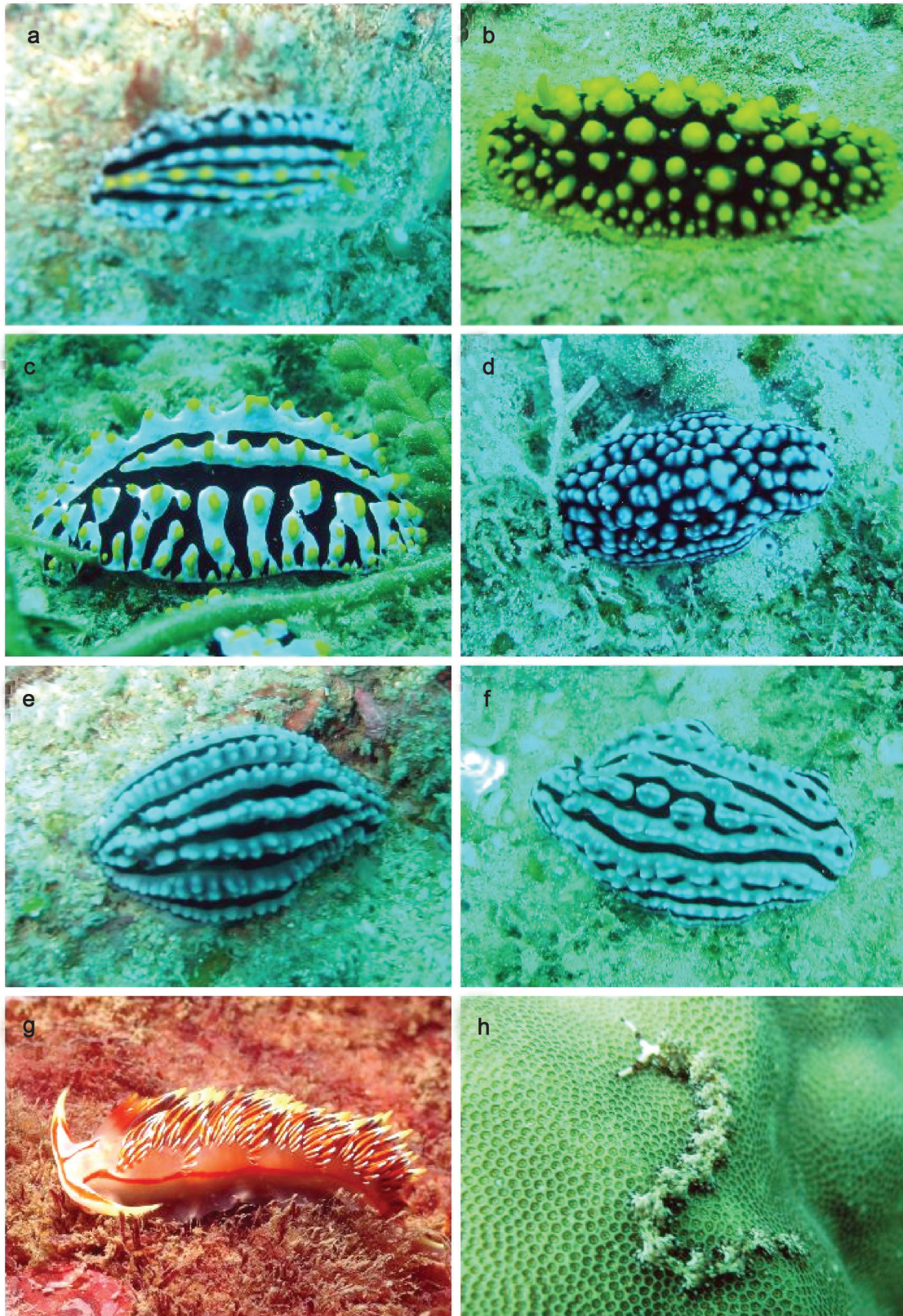


Fig. 3. Underwater photographs of the species observed (a) *Phyllidia alyta*, (b) *Phyllidia ocellata*, (c) *Phyllidia varicosa*, (d) *Phyllidiella pustulosa*, (e) *Phyllidiella rosans*, (f) *Phyllidiella zeylanica*, (g) *Phidiana militaris*, (h) *Pteraeolidia ianthina*

of the mantle, one on each side of a single median blue ridge. The median ridge has a single row of yellow capped tubercles. There are also many black rays or lines, on either side of the mantle, running at right angles to the mantle edge. The ventral surface of mantle and the foot are grey (Fig. 2f).

Material observed: Kariyachalli Island (08° 57' 02.6", 078° 15' 18.8"); maximum length of the crawling animal: ~6 cm; depth: 15 m; reef area with rubbles and pebbles along with shells.

Distribution: Indian Ocean from northern Western Australia to eastern Africa.

Remarks: New record to Gulf of Mannar Biosphere Reserve

F. marindica appears to be closest to *F. menindie* which is found in the western Pacific Ocean. This species has often been confused with *F. ruppelii* Bergh, which is considered by Brunckhorst to be confined to the Red Sea. It is also likely to be mistaken for the widespread *Phyllidia varicosa* but in that species there are always three or five longitudinal ridges on the dorsum and the anus opens in the dorsal midline.

Phyllidia alyta Yonow, 1996

Description: This species is characterised by four longitudinal black bands which rarely meet. The inner pair runs from behind the rhinophores to just in front of the anal papilla. Between them is the median row of tubercles which are usually tipped with yellow. There is a second row of tubercles on each side which are rarely tipped with yellow. Outside this row is a second black line on each side which runs from the anterior edge of the mantle back to behind the anal papilla. There is also usually a black transverse band between the rhinophores. Outside the longitudinal black lines is another row of tubercles and outside rows there are small irregularly scattered tubercles at the mantle edge. There is a solid black median line on the sole of the foot which is broken anteriorly into one or two patches. The rhinophores are yellow (Fig. 3a).

Material observed: Nallathanni Island (09° 06' 52.9", 078° 34' 44.7"); maximum length of the crawling animal: ~6 cm; depth: 12 m; rocky area covered with algae; Vilankuchalli Island (08° 56' 21.9", 078° 16' 17.5"); maximum length of the crawling animal: ~6 cm; depth: 15 m; rocky area covered with algal mat.

Distribution: Western Indian Ocean (Maldives, Mauritius, Myanmar and India)

Remarks: New record to Gulf of Mannar Biosphere Reserve

Phyllidia ocellata Cuvier, 1804

Description: The most typical dorsal pattern consists of four to ten black rings bordered in white, each with a gold

coloured centre. There is little ontogenetic variation in external pattern (*i.e.*, juveniles resemble adults), however considerable individual variation in dorsal pattern occurs in this species. From the basic pattern described, there appear to be numerous related and irregular patterns where the black areas expand and/or meander across the dorsum (Fig. 3b).

Material observed: Kariyachalli Island (08° 57' 02.6", 078° 15' 18.8"); maximum length of the crawling animal: ~5 cm; depth: 12 m; sandy area; "Mainland" (09° 02' 39.2", 078° 30' 20.4"); maximum length of the crawling animal: ~8 cm; depth: 20 m; sandy area covered with seaweeds Nallathanni Island (09° 06' 31.9", 078° 34' 29.6"); maximum length of the crawling animal: ~6 cm; depth: 15 m; found on a sponge.

Distribution: Common throughout the tropical Indo-West Pacific Ocean, including the Red Sea.

Remarks: Reported earlier as *Phyllidia multituberculata* by O'donoghue (1932).

Phyllidia varicosa Lamarck, 1801

Description: The body is firm, 3-6 longitudinal tuberculate notal ridges on the dorsum. The ridge and bases of the tubercles are blue-grey in colour and the tubercles are capped in yellow. The foot sole possesses a black longitudinal foot stripe. This species also have close resemblances with *P. coelestis* which has a brown black band down the midline while in *P. varicosa* there is a bluish grey ridge down the middle line. This species is also confused with *P. alyta* which possesses a solid black median line in the sole of the foot (Fig. 3c).

Material observed: "Mainland" (08° 45' 30.0", 78° 14' 20.0"; 09° 04' 00.3", 078° 35' 50.0"); maximum length of the crawling animal: 10 - 12 cm; depth: 15 m; sandy area covered with seaweeds.

Distribution: Throughout the Indo-West Pacific Ocean including the central Pacific and the Red Sea.

Remarks: New record to Gulf of Mannar Biosphere Reserve

Phyllidiella pustulosa (Cuvier, 1804)

Description: *P. pustulosa* has three median clusters of pink tubercles. The tubercles are in a cluster which is separated in adult specimen; however it is amalgamated in the juvenile stage. The intensity of pink colouration is possibly related to diet and time since feeding. The mantle has a pale pink edge, the broad and oral tentacles are triangular and black tipped (Fig. 3d).

Material observed: Upputhanni Island (09° 05' 09.1", 078° 29' 37.9"); maximum length of the crawling animal: ~7 - 10 cm; depth: 18-20 m; sandy area covered with seaweeds.

Distribution: Throughout the tropical Indo-West Pacific.

Remarks: New record to Gulf of Mannar Biosphere Reserve

Phyllidiella rosans (Bergh, 1873)

Description: *Phyllidiella rosans* is distinguished by its numerous, longitudinal, rounded pink ridges, pink margin and black rhinophores which have a pink stalk. Ventrally, *P. rosans* is pale grey. The longitudinal notal ridges of the present species distinguish it from the similar species of *Phyllidiella*. The only other species of *Phyllidiella* which possesses longitudinal ridges and a pale foot is *P. zeylanica*. However, that species differs from *P. rosans* in the very uneven, irregular nature of the notal ridges which are formed from the coalesced bases of compound tubercles. Furthermore, the foot of *P. zeylanica* is whitish and its gills and oral tentacles are dark grey (pale in *P. rosans*) (Fig. 3e).

Material observed: Kariyachalli Island (08° 57' 02.6", 078° 15' 18.8"); maximum length of the crawling animal: ~7 cm; depth: 10 m; rocky area covered with algal mat and seaweeds.

Distribution: Western Pacific Ocean and the Indian Ocean (Maldives, East Africa and India)

Remarks: New record to Gulf of Mannar Biosphere Reserve

Phyllidiella zeylanica (Kelaart, 1859)

Description: The species is characterized by the presence of three black concentric bands running round the notum and an interrupted median dark band. The median band and the first concentric dark band close to the former are prominent. The concentric band is incomplete posteriorly leaving a small gap. The other two more marginally situated bands are very narrow and inconspicuous. In between the dark concentric bands are prominently elevated greenish yellow longitudinal ridges bearing tubercles of varying sizes. These ridges run across in front of the rhinopore and behind the anal region. Mantle tubercular, salmon-coloured; three continuous black lines run round the whole length; the internal one broader, taking within its circuit the dorsal tentacles and anal orifice; two other lines run parallel to this all round the mantle, the outer one narrowest. Dorsal tentacles are large, conical, pointed and circularly laminated at the upper half, which is black in colour (Fig. 3f).

Material observed: Mainland (08° 49' 28.8", 078° 14' 28.62"); maximum length of the crawling animal: ~6 cm; depth: 15 m; sandy area covered with algae.

Distribution: Tropical Indian Ocean (Eastern Africa to Java)

Remarks: New record to Gulf of Mannar Biosphere Reserve

ORDER : NUDIBRANCHIA Blainville, 1814
SUB-ORDER : AEOLIDINA Odhner, 1934
FAMILY : GLAUCIDAE Oken, 1815

Phidiana militaris (Alder and Hancock, 1864)

Description: Body rather stout, tapering to a fine point behind, of a pale flesh colour or nearly white. Dorsal tentacles rather long, smooth, swelling a little in the centre, and tapering to a point above, with a broad belt of scarlet near the centre, gradually fading into flesh colour below. Bright yellow on the upper half, excepting the extreme tips, which are white. Oral tentacles large, stout, and tapering, color as the dorsal, excepting that the scarlet central belt is extended down each tentacle in narrow line, the two uniting on a bulb on the central of the head, and continued thence backwards to the commencement of the branchial papillae. Branchial papillae numerous, moderately stout, of a reddish brown colour, with bright yellow tips, forming six clusters on each side, and nearly meeting on the back (Fig. 3g).

Material observed: Kariyachalli Island (08° 57' 25.5", 078° 15' 21.3"); maximum length of the crawling animal: ~13 cm; depth: 16 m; rocky area covered with algae.

Distribution: Gulf of Oman, Thailand, Hong Kong, Philippines and India.

Remarks: Reported four times from this area as *Caloria militaris* and *Hervia militaris*.

Pteraeolidia ianthina (Angas, 1864)

Description: The mantle greyish in colour; the colour varies from place to place. Distinguished by the length and branched ceratal group at particular interval. A greenish white stripe from the anterior of the mantle to the end. *Pteraeolidia* stores zooxanthellae in its body, which helps them to stay without taking food for few days. The white animal is a juvenile usually found in lush growths of short turfing hydroids (Fig. 3h).

Material observed: "Mainland" (08° 45' 30.0", 078° 14' 20.0"); maximum length of the crawling animal: ~15 cm; depth: 23 m; sandy area covered with seaweeds.

Distribution: Throughout the tropical and subtropical Indo-West Pacific.

Remarks: New record to Gulf of Mannar Biosphere Reserve. Studies on the nudibranch fauna from Gulf of Mannar area is very scant compared to the global scenario. Since no previous works were carried out from the Vembar and Tuticorin group of Islands, they remain unexplored. In the light of the biodiversity conservation activities in the area, greater attention is needed for the reef associated fauna of these coral reefs. Several coral reef areas that once thrived has been exploited for industrial purpose and virtually little exist today, particularly from Tuticorin to Krusadai Islands (Patterson *et al.*, 2004). If comprehensive studies are not carried out at this area, several species may become extinct even before their mere existence is noticed.

Acknowledgements

Authors wish to thank Sri. Senbagamoorthy, Department of Forest, Ramnad District; Mr. Malkani, Director, Mr. Naganathan IFS, Eco Development Officer, Gulf of Mannar Biosphere Reserve Trust for the help and permission accorded to survey the Islands. Authors also wish to express their sincere gratitude to the staff of Peoples Action for Development, NGO, Vembar especially to the Executive Director Mr. Rajendraprasad for the help in field surveys. Due gratitude is also expressed to the Principal and staff members of Kamaraj college, Thoothukkudi for the help rendered.

References

- Alder, J. and Hancock, A. 1864. Notice of a collection of nudibranchiate Mollusca made in India by Walter Elliot Esq., with descriptions of several new genera and species. *Trans. Zool. Soc. Lond.*, 5: 113-147, pls. 28-33.
- Eliot, C. 1906. On the nudibranchs of South India and Ceylon, with special reference to the drawings by Kelaart and the collection belonging to Alder and Hancock preserved in the Hancock museum at New Castle-on-Tyne North. *Proc. Zool. Soc. Lond.*, 1906 pt. II: 636-691 and 997-1008.
- Farran, G. P. 1905. Report on the Opisthobranchiate Mollusca collected by Professor Herdman at Ceylon in 1902. *Ceylon Pearl Oyster Fisheries Supp. Report*.
- Fontana, A., Maria Ietizia Ciavatta, Lisette D'souza, Ernesto Mollo, Chandra G. Naikk, Parameswaran, P. S., Solimabi Wahidulla and Guido Cimino. 2001. Selected chemo-ecological studies of marine opisthobranchs from Indian coasts. *J. Indian Insti. Sci.*, 81(4): 403-415.
- Gravely, F. H. 1942. Shells and other animal remains found on the Madras beach. II. Snails, etc. (Mollusca Gastropoda). *Bulletin of the Madras Government Museum, New Series, Natural History Section* 5(2): 1-110.
- O'donoghue, C. H. 1932. Notes on Nudibranchiata from southern India. *Proc. of the Malacological Society of London*. 20: 141-166.
- Patterson Edward, J. K., Jamila Patterson., Venkatesh, M., Mathews. G., Chellaram, C. and Dan Wilhelmsson. 2004. A field guide to stony corals (Scleractinia) of Tuticorin in Gulf of Mannar, Southeast India. *SDMRI Special Research Publication* No. 4: 80 pp.
- Pillai, C. S. G. 1971. Composition of the coral fauna of the southeast coast India and the Laccadives. *Symp. Zool. Soc. London*. 28: 301-327.
- Pillai, C. S. G. 1977. The structure, formation and species diversity of South Indian reefs. *Proc. 3rd Inter. Coral reef Sym.*, Miami. Vol. 1. 41-53.
- Rao, K. P. 1968. On a new genus and some new species of opisthobranchiate gastropods of the family Eubranthidae from the Gulf of Mannar. *Proceedings of the Symposium on Mollusca* Cochin, January 12 to 16, 1968. *Symposium series* 3, part 1, Marine Biological Association of India, Mandapam Camp, India, pp. 51-60.
- Rao, K. V. 1962a. [For 1961]. Development and life history of a nudibranchiate gastropod, *Cuthona adyarensis* Rao. *J. Mar. Biol. Ass. India.*, 3: 186-197.
- Rao, K. V. 1962b. [For 1961]. On two opisthobranchiate molluscs, *Placobranchus ocellatus* Hasselt and *Discodoris boholiensis* Bergh, from Indian waters not hitherto been recorded. *J. Mar. Biol. Ass. India.*, 3(1, 2): 253-256.
- Rao, K. V. 1967a. On the structure and life history of a new aeolid *Favorinus argentimaculatus* from Palk Bay. *Advance Abstracts Contributions Fisheries and Aquatic Science India.*, 1(4): 41-42.
- Rao, K. V. 1967b. On a few little known opisthobranchiate Mollusca from the Palk Bay and the Gulf of Mannar with notes on their development (Abstracts), *Contributions Fisheries and Aquatic Science India*, 1(4): 51.
- Rao, K. V. 1968. On the structure and life-history of a new aeolid, *Favorinus argentimaculatus* from Palk Bay. *Symposium on Mollusca* Marine Biological Association of India, Cochin, 12-16 January 1968, p. 7.
- Rao, K. V. 1973. On dendronotacean nudibranch gastropods of the genera *Hancockia* and *Doto* from the Gulf of Mannar. *Special publication* Marine Biological Association of India, Cochin India, 1973, p. 321-332.
- Rao, K. V. and Alagaraswami, K. 1961. External morphology and early development of *Pleurobranchus (Oscanus)* sp. from the Gulf of Mannar. *J. Mar. Biol. Ass. India.*, 3 (1 and 2).
- Rao, K. V. and Alagarswami, K. 1960. An account of the structure and early development of a new species of a nudibranchiate gastropod, *Eolidina (Eolidina) mannarensis*. *J. Mar. Biol. Ass. India.*, 2: 6-16.
- Rao, K. V. and KrishnaKumari, L. 1973. Structure and early development of an opisthobranch mollusc, *Caloria militaris* (Alder & Hancock). *Indian J. Mar. Sci.*, 2(1): 32-37.
- Rao, K. V. and KrishnaKumari, L. 1974. On some aspects of taxonomy, structure and early development of the nudibranchiate gastropod, *Discodoris fragilis* (Alder & Hancock). *J. Mar. Biol. Ass. India.*, 16(3): 689-699.
- Rao, K. V. and Rao, K. P. 1963. *Stiliger nigrovittatus* sp. nov, a sacoglossan mollusc from the Gulf of Mannar. *J. Mar. Biol. Ass. India.*, 5(2): 232-238.
- Reddiah, K. 1970. The topography of Appa Island and its fringing reef in the Gulf of Mannar. *Proceedings Indian Science Congress Association*, 57(4): 405-406.
- Rudman, W. B. 1987. The Chromodorididae (Opisthobranchia: Mollusca) of the Indo-West Pacific: *Chromodoris epicuria*, *C. aureopurpurea*, *C. annulata*, *C. coi* and *Risbecia tryoni* colour groups. *Zool. J. Linnean Soc.*, 90: 305-407.

- Satyamurti, S. T. 1952. The Mollusca of Krusadai Island (in the Gulf of Mannar). New Series, Natural History Section, *Bulletin of Madras Government Museum*, 1(2) pt. 6, p. 1-258, pls. 1-34.
- Valdes, A., Ernesto Mollo, and Ortea, J. 1999. Two new species of *Chromodoris* (Mollusca, Nudibranchia, Chromodorididae) from southern India with a redescription of *Chromodoris trimarginata* (Winckworth, 1946). *Proceedings of the California Academy of Sciences*, 51(13): 461- 472.